TSA Meeting — Slide 9 Study Note

Ch 4: SAMBP-SyncOC — Adaptive Generator Over-Current Protection

Contributions C2 · Chapter 4 · SAMBP Framework

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Slide Overview and Narrative

What This Slide Shows

Slide 9 presents **Ch 4: SAMBP-SyncOC**, the adaptive generator overcurrent protection framework. The slide's narrative has three parts:

The problem: Synchronous generators (SGs) operating near IBR buses experience *depressed fault current*. When IBRs current-limit during a fault, they reduce the total short-circuit contribution, compressing the fault current seen by the generator's OC relay. Settings calibrated for an all-SG network (where fault current is high) become either too sensitive (causing spurious pickups on load transients) or too slow (failing to trip high-impedance faults). Additionally, IBR reactive injection alters the time-overcurrent coordination between primary and backup relays.

The solution: SAMBP-SyncOC replaces the fixed pickup I_p and time-multiplier TMS with an **online adaptive setting** derived from the LM inverse estimator. The pickup $I_{\text{set}}(k_{\text{ibr}})$ is updated every fault cycle using the estimated steady-state fault current $\hat{I}_{ss}$, which already accounts for the actual IBR penetration at the time of the fault. The coordination constraint $t_{\text{primary}} + \text{CTI} \leq t_{\text{backup}}$ is maintained for all k_{ibr} via a bounded projected-gradient update — i.e., every setting change is simultaneously tested against the CTI constraint and clipped if it would violate coordination.

The result: The LM model fits the measured fault current waveform with $R^2 =$ 0.999+ (versus 0.941 for the fixed offline curve). Spurious pickups fall from 38 to 16 (-58%). Coordination margin improves from 0.82 to 0.94. Maloperation under IBR penetration drops from 12.3% to 1.1%.

Slide Key Results

Metric	Fixed Settings	SAMBP-SyncOC	Change
R^2 (OC curve fit)	0.941	>0.999	+6.2% relative
Spurious pickups	38	16	−58%
Coordination margin	0.82	0.94	+0.12
Maloperation under IBR	12.3%	1.1%	−91% relative

Key observation from slide: Online $I_{\text{set}}(k_{\text{ibr}})$ adaptation cuts spurious pickups by 58% while maintaining IEC 60255-151 coordination. The R^2 jump from 0.941 to 0.999 means the model is now characterising the actual fault waveform — not a generic worst-case approximation.

Engineering Challenge: Why Fixed OC Settings Fail Near IBR Buses

Four Failure Modes of Offline-Set OC Relays in IBR Networks

Classical OC protection sets I_p and TMS at commissioning using worst-case fault current calculations. This approach fails in four scenarios that routinely arise in modern Active Distribution Networks (ADNs):

1. **Reconfigurable networks:** Post-fault topology changes alter the Thevenin impedance seen by the generator, shifting fault current by 20–80%. A commissioning-time setting is immediately wrong after any switching event.
2. **Variable loading:** A pickup calibrated for peak load ($I_{\text{load}}^{\max}$) is far above off-peak load. During off-peak, a high-impedance fault produces only slightly more current than load — the relay cannot distinguish them.
3. **IBR fault current suppression:** IBRs current-limit to 1.0–1.2 pu of rated during fault ride-through. In an IBR-dominated network, the total fault current may be only 1.5–2.0 pu, well below the OC pickup set for an all-SG system (2.5–5 pu). The relay does not pick up, and the fault persists.
4. **Generator aging:** Machine parameters (X_d'', T_d'', R_a) drift over service life. Settings based on nameplate values become stale within 3–5 years, yet recalibration requires an outage.

The most immediately damaging in an IBR-heavy network is item 3: the combined fault current is so depressed that the fixed pickup threshold is never reached for high-impedance faults, causing **maloperation rate 12.3%** with fixed settings.

What "Depressed Fault Current" Means Quantitatively

For a three-phase fault at the generator bus in a mixed SG-IBR network:

$$I_{\text{fault,total}} = \underbrace{\frac{E_q''}{X_d'' + X_{th}}}_{\text{SG contribution}} + \underbrace{k_{\text{ibr}} \cdot I_{\text{rated}}}_{\text{IBR (limited)}}$$

For $X_d'' = 0.20$ pu, $X_{th} = 0.15$ pu, $E_q'' \approx 1.0$ pu: SG contribution $= 1.0/0.35 = 2.86$ pu. IBR contributes $k_{\text{ibr}} \times 1.2$ pu (limited to 120% rated). Total: $2.86 + 0.5 \times 1.2 = 3.46$ pu at $k_{\text{ibr}} = 0.5$. Compare with all-SG equivalent: 5.0 pu. The fault current is **reduced by 31%** — and this is for a bolted three-phase fault. For a single-line-to-ground (SLG) fault at $R_F = 10\,\Omega$, the reduction is 55–65%.

Reduced-Order Fault Current Model (C6)

Six-Parameter SG Fault Current Model

The phase-A fault current is modelled as the superposition of three components:

$$\hat{i}_a(t;\boldsymbol{\theta}) = \underbrace{I_{ss}}_{\text{steady-state}} + \underbrace{I_{sub}\, e^{-(t-t_0)/\tau_{ac}} \cos(\omega_0 t + \phi_a)}_{\text{subtransient AC}} + \underbrace{I_{DC}(t_0)\, e^{-(t-t_0)/\tau_{dc}}}_{\text{DC offset}} \quad (1)$$

with parameter vector $\boldsymbol{\theta} = \{t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a\}$.

The DC amplitude is *not* a free parameter; it is fixed by the continuity condition at $t = t_0$:

$$I_{DC}(t_0) = -I_{sub}\cos\phi_a - I_{ss} \tag{2}$$

This constraint reduces the formulation from seven to six free parameters, which is the key numerical contribution of C6. The Jacobian condition number drops from $\kappa_n \approx 2100$–5800 (seven parameters) to $\kappa_n = 8.3$–18.4 (six parameters) — a **250–315× reduction**. This makes the LM optimisation converge reliably without ill-conditioning.

Two-Pass LM Estimation Strategy

The time-scale separation $\tau_{ac} \approx T''_d = 30$ ms $\ll \tau_{dc} \approx T_a = 100$ ms $\ll T'_d = 800$ ms enables a two-pass decomposition:

Pass	Window	Free parameters	Fixed	Purpose
Pass 1	$[t_0,\ t_0 + 60\,\text{ms}]$	$t_0, I_{sub}, \tau_{ac}, \phi_a, \tau_{dc}$	$I_{ss} = I_{\text{load}}$	Subtransient fit
Pass 2	$[t_0,\ t_0 + 250\,\text{ms}]$	I_{ss}	Pass-1 values	Steady-state tail

Pass 1 captures the fast subtransient ($\tau_{ac} \approx 30$ ms). Pass 2 uses the long tail to separate the steady-state I_{ss} from the decaying subtransient. Stopping criterion: $\|\mathbf{r}\|_2^2/K < 10^{-6}$ pu^2.

From the paper figure (fig_key_results): The waveform fit panel shows the measured phase-A fault current (blue), the six-parameter model (red, R^2=0.999), the envelope (orange dotted), and the DC offset (green dashed) with identified parameters $\hat{I}_{sub} = 2.838$ pu, $\hat{\tau}_{dc} = 55.9$ ms, $\hat{\phi}_a = \pi/2$ rad.

Four-Component Confidence Score and Adaptation Gate (C7)

Why a Confidence Gate is Essential

The LM estimator will always produce a result — even for a noisy measurement or a CT saturation event where the fault current waveform does not match the six-parameter model. Without a quality gate, SAMBP-SyncOC would blindly update relay settings based on a poor-quality estimate. Contribution C7 introduces a **composite confidence score** $\gamma \in [0, 1]$ that gates the adaptation decision:

$$\gamma = w_1\gamma_R + w_2\gamma_J + w_3\gamma_\Delta + w_4\gamma_P$$

with weights $w_1 = 0.35$, $w_2 = 0.30$, $w_3 = 0.20$, $w_4 = 0.15$ (sum $= 1.0$).

Component	Symbol	Measures	Good value
Residual fit	γ_R	R^2 of waveform fit; sharpness 100	$\gamma_R \to 1$ if $R^2 \to 1$
Jacobian condition	γ_J	$\exp(-\alpha_J\kappa_n)$	$\gamma_J \to 1$ if κ_n low
Parameter stability	γ_Δ	Change in $\boldsymbol{\theta}$ from last estimate	$\gamma_\Delta \to 1$ if stable
Physical plausibility	γ_P	$\boldsymbol{\theta} \in \Theta$ (parameter bounds)	1 if all in bounds, 0 otherwise

Gate: If $\gamma \geq \gamma_{\text{th}} = 0.70 \Rightarrow$ adaptation authorised; else relay holds previous settings.
Validated: $\gamma \in [0.89, 0.93]$ for all three fault cases — comfortably above gate.

Adaptive Setting Update Law and Coordination Constraint (C8)

IEC 60255-151 Very-Inverse Relay with Adaptive Pickup

The slide's inverse-time equation in standard IEC form is:

$$t_{\text{op}} = \frac{\text{TMS} \cdot A}{(I/I_{\text{set}}(k_{\text{ibr}}))^{B} - 1} \tag{3}$$

For the IEC Very-Inverse curve: $A = 13.5$, $B = 2$. For Standard Inverse: $A = 0.14$, $B = 0.02$. For Extremely Inverse: $A = 80$, $B = 2$. The thesis uses Very-Inverse throughout.

The adaptive pickup is derived from the LM-estimated steady-state fault current:

$$I_p^{\text{adapt}} = k_s \cdot I_{ss}^{\text{load}} \cdot (1 + \delta_p) \tag{4}$$

where $k_s = 1.25$ (sensitivity factor ensuring trip for all fault levels above load) and $\delta_p = 0.10$ (10% security margin).

The adaptive TMS is solved directly from the coordination constraint:

$$\text{TMS}^{\text{adapt}} = \frac{t_{\text{CTI}} \cdot [(I_{\text{fault}}^{\text{max}}/I_p)^2 - 1]}{k_{\text{VI}}} \tag{5}$$

Coordination Constraint: t-primary + CTI less than t-backup

The coordination time interval (CTI) constraint requires that the primary relay trips before the backup relay by a margin CTI ≥ 0.3 s (IEC 60255-151 minimum):

$$t_{\text{primary}}(I_p, \text{TMS}) + \text{CTI} \leq t_{\text{backup}}(I_p^{\text{backup}}, \text{TMS}^{\text{backup}})$$

When I_p is updated, TMS must be simultaneously adjusted to maintain this inequality for *all* values of $k_{\text{ibr}} \in [0, k_{\text{ibr}}^{\text{max}}]$.

Projected-Gradient Update Mechanism

The "projected-gradient optimisation" mentioned on the slide is the bounded update law:

1. **Gradient step:** Compute the unconstrained optimal I_p from the LM estimate.
2. **Project onto feasible set:** Clip to the safety bounds:

$$I_p \leftarrow \text{clip}(I_p,\ I_{p,\min},\ I_{p,\max}) \tag{6}$$
$$I_{p,\min} = 1.05 \cdot I_{\text{load}}^{\max} \quad \text{(never trips on pure load)} \tag{7}$$
$$I_{p,\max} = 0.80 \cdot I_{\text{fault,3P}}^{\min} \quad \text{(always trips bolted fault)} \tag{8}$$
$$|\Delta I_p| \leq \delta_{\max} = 0.10 \cdot I_{p,0} \quad \text{(max 10\% per cycle)} \tag{9}$$

3. **Check coordination:** Verify $t_{\text{primary}}^{\text{new}} + \text{CTI} \leq t_{\text{backup}}$ for the worst-case k_{ibr}. If violated, scale back I_p toward $I_{p,0}$ until constraint is satisfied (secondary projection).
4. **Authorise:** Only execute if $\gamma \geq \gamma_{\text{th}} = 0.70$ (C7 gate).

The security property guaranteed by this law (Proposition from Ch 2): if $I_{p,0}$ trips all bolted faults, then any adapted $I_p \leq I_{p,\max}$ also trips all bolted faults — **trip security is preserved under bounded adaptation**.

Results: What the Numbers Mean

Interpreting the Four Metrics

R^2 = 0.941 (fixed) vs 0.999 (adaptive): The fixed relay's OC curve was calibrated with worst-case fault current magnitudes from a system study, not a fitted waveform model. $R^2 = 0.941$ means the standard IEC characteristic describes only 94.1% of the variance in actual fault current behaviour — the remaining 5.9% includes IBR-induced distortions (harmonic injection, reactive current transients). SAMBP-SyncOC fits the actual measured waveform, achieving $R^2 = 0.999+$.

Spurious pickups: 38 → 16 (−58%): A "spurious pickup" is when the relay picks up (current exceeds I_p) during a load transient, motor start, or IBR reactive injection event — not a genuine fault. With fixed $I_p = 1.20$ pu set for an all-SG network, even moderate off-peak load transients cross the threshold. With $I_p^{\text{adapt}} = 1.49$ pu (adapted to the actual IBR-depressed fault current and load level), the threshold is higher relative to load transients, reducing false pickups by 58%.

Coordination margin: 0.82 → 0.94: Coordination margin $= t_{\text{backup}}/t_{\text{primary}}$ at the maximum fault current. A margin of 0.82 means that at maximum fault current, the primary relay operates at 82% of the backup relay time — only an 18% margin. If the primary relay is slightly slow due to CT burden, this margin could be violated. SAMBP-SyncOC actively adjusts TMS to maintain a 0.94 ratio across all k_{ibr} values.

Maloperation under IBR: 12.3% → 1.1%: A "maloperation" is either a missed trip (fault not detected) or a spurious trip (non-fault condition). With fixed settings, 12.3% of IBR-modified fault events result in maloperation — primarily missed trips on high-impedance faults where the IBR-depressed fault current stays below $I_p = 1.20$ pu. With adaptive settings tracking the actual IBR level, maloperation drops to 1.1%.

From the Paper Figures: Relay Characteristic

The relay characteristic figure (fig_relay_characteristic, from the SyncOC paper) shows the IEC Very-Inverse log-log plot with:

- **Blue solid**: Fixed relay ($I_p = 1.20$ pu, TMS = 0.050).
- **Adaptive**: $I_p = 1.487$ pu, TMS = 0.050 (same TMS, higher pickup).
- **Blue circle**: Operating point for 3PH bolted fault with adaptive settings — faster trip.
- **Orange square**: Operating point for SLG fault ($R_F = 0.15$ pu) with fixed settings — this falls *near* the fixed pickup (close to "no trip" boundary).
- **Orange shaded band**: "Security margin gain" — the region between the fixed and adaptive pickup thresholds where the relay is now desensitised to load transients (fewer spurious pickups).

The key insight: raising I_p from 1.20 to 1.49 pu does not slow the relay on genuine high-current faults (the Very-Inverse curve is steep there), but it eliminates the spurious pickup region between 1.20 and 1.49 pu where only load transients live.

Best Figure to Show at the TSA Meeting

Recommended Figure: fig_key_results — Two-Panel Waveform and Relay Response

What it shows:

- **Left panel**: Phase-A fault current waveform. The measured waveform (blue) is overlaid with the six-parameter LM model (red, $R^2 = 0.999$). Orange dotted lines show the AC envelope; green dashed line shows the DC offset component. Identified parameters in the legend box: $\hat{I}_{sub} = 2.838$ pu, $\hat{\tau}_{dc} = 55.9$ ms, $\hat{\phi}_a = \pi/2$ rad.
- **Right panel**: Adaptive relay response. Blue curve = I_{rms} after fault inception. The fixed pickup $I_p^{\text{fix}} = 1.20$ pu (dash-dot) is below the decaying fault current for a long time, causing slow trip. The adaptive pickup $I_p^{\text{adapt}} = 1.49$ pu (red shaded band) is set precisely at the right level: the relay picks up at the correct moment (dashed trip line at ≈ 90 ms) and the security margin gain (pink band) shows the elevated threshold that eliminates spurious pickups.

Why it is the best figure:

1. It directly demonstrates both contributions: estimation accuracy (left) and relay improvement (right).
2. A TSA examiner can immediately see the $R^2 = 0.999$ label on the waveform.
3. The security margin gain band is visually intuitive — it shows exactly why raising I_p reduces spurious pickups.
4. It connects the abstract LM algorithm to a concrete engineering outcome in one figure.

Where to find it: `sambp/sambp_sync_oc/paper/figures/fig_key_results.pdf`
Secondary figure: `fig_relay_characteristic.pdf` (IEC Very-Inverse log-log plot with fixed vs. adaptive operating points). Use this to explain the coordination constraint — point to the operating point markers and show how the CTI margin is maintained.
Whiteboard supplement: Draw three horizontal lines: (1) $I_{\text{load}}^{\max}$ at the bottom, (2) $I_p^{\text{fix}} = 1.20$ pu slightly above it, (3) $I_p^{\text{adapt}} = 1.49$ pu higher still. Annotate the gap between (1) and (2) as "spurious pickup zone" and the gap between (2) and (3) as "security margin gain". This takes 30 seconds and makes the adaptive pickup concept unmistakable.

Cross-Thesis Connections

How This Chapter Connects to Other Work

Depends on (upstream):

- **Ch 2 (SAMBP Foundations)**: The two-pass LM algorithm, confidence score framework, and bounded update law are all developed in Ch 2 and applied here. The "projected-gradient" name comes from the Ch 2 formalisation.
- **Ch 3 (IBR Characterisation)**: The depressed fault current mechanism (IBR current limiting during FRT) is characterised as failure modes F1–F3 in Ch 3. This chapter solves F1 (fixed OC insensitive to IBR-level fault current).

- **Ch 6 (Digital Twin / EKF)**: The real-time $\hat{k_{\text{ibr}}}$ estimate from the EKF is used in Eq. (4) to track which operating regime the network is in.

Enables (downstream):

- **Ch 7 (WAPC)**: The CTI constraint $t_{\text{primary}} + \text{CTI} \leq t_{\text{backup}}$ from this chapter is the inter-relay coordination constraint that the Wide-Area Protection Coordinator (WAPC) enforces at the network level. WAPC extends the bilateral CTI of SyncOC to a full N-relay coordination optimisation.
- **Planned C9 (HIF Extension)**: The adaptive sensitivity boundary $R_F^{\max}(k_{\text{ibr}})$ extends the framework to high-impedance fault detection. Expected to reduce HIF miss rate by 30–50%.
- **Planned C10 (Park dq validation)**: The six-parameter reduced model (Eq. (1)) will be validated against the full 6th-order Park dq truth model to confirm the approximation error $< 5\%$ RMS for $t \leq 250$ ms.

Confidence score context: The γ framework from C7 is the *same architectural pattern* as the Stage-2 veto gate in SAMBP-87T, 87L, and 87B. SAMBP-SyncOC is the first application where the Stage-2 gate directly modifies relay settings (not just vetoes a trip decision). This generalisation — using Stage-2 confidence to adapt parameters, not just to block/permit — is a novel architectural contribution of this thesis.

Key Equations Reference Card

Equations You Must Be Able to Write and Explain

1. Fault current model (slide core equation):

$$\hat{i}_a(t;\boldsymbol{\theta}) = I_{ss} + I_{sub}\, e^{-(t-t_0)/\tau_{ac}} \cos(\omega_0 t + \phi_a) + I_{DC}(t_0)\, e^{-(t-t_0)/\tau_{dc}}$$

Say: "Three components: steady-state, subtransient AC, DC offset. Six free parameters. The DC amplitude is not free — it is set by continuity to eliminate the 7th parameter."

2. DC continuity constraint (the key reduction step):

$$I_{DC}(t_0) = -I_{sub} \cos\phi_a - I_{ss}$$

Say: "At $t = t_0$, the current must be continuous. This fixes I_{DC} in terms of I_{sub} and I_{ss}, eliminating one free parameter and reducing the Jacobian condition number by 250–315×."

3. Adaptive inverse-time OC relay (slide equation):

$$t_{\text{op}} = \frac{\text{TMS} \cdot A}{(I/I_{\text{set}}(k_{\text{ibr}}))^B - 1}, \quad A = 13.5,\ B = 2 \text{ (IEC Very-Inverse)}$$

Say: "The only change from a classical IEC relay is that I_p (the pickup) is updated online. TMS and the curve constants A, B remain fixed. The curve shape is the same; the threshold moves."

4. Adaptive pickup setting:

$$I_p^{\text{adapt}} = k_s \cdot \hat{I}_{ss}^{\text{load}} \cdot (1 + \delta_p) = 1.25 \times \hat{I}_{ss} \times 1.10$$

Say: "The LM estimator gives us $\hat{I}_{ss}$, the steady-state load current under the current network state. We add 25% sensitivity margin and 10% security margin. The result is an I_p that tracks the actual operating conditions."

5. Coordination constraint (slide equation):

$$t_{\text{primary}}(I_p, \text{TMS}) + \text{CTI} \le t_{\text{backup}}, \quad \text{CTI} \ge 0.3\,\text{s (IEC 60255-151)}$$

Say: "This is the standard selectivity requirement: the primary relay must trip before the backup, with a minimum margin of 300 ms to allow for breaker operating time. SAMBP verifies this for all k_{ibr} before applying any setting change."

6. Confidence gate:

$$\gamma = 0.35\gamma_R + 0.30\gamma_J + 0.20\gamma_\Delta + 0.15\gamma_P \ge 0.70$$

Say: "Four quality checks: fit quality, Jacobian condition, parameter stability, physical plausibility. All must collectively score above 70% before the relay settings are updated."

Speaker Script (Timed to 3 Minutes)

What to Say at the TSA Meeting

[0:00–0:35] Open with the problem:
"Overcurrent protection of synchronous generators has been standardised for decades — you set a pickup current and a time-multiplier at commissioning and never touch them again. That works when the fault current is predictable. But in an IBR-dominated network, it is not. IBRs limit their fault current to 1.0–1.2 times rated during fault ride-through. In a network with $k_{\text{ibr}} = 0.5$, the total short-circuit current can be reduced by 30–60%. A pickup set for an all-synchronous-generator network may never be reached for a genuine high-impedance fault. That is how you get a 12.3% maloperation rate — and that is exactly what we measured with fixed settings."

[0:35–1:20] Explain the solution:
"SAMBP-SyncOC replaces the fixed pickup with an adaptive one. After each fault, the LM inverse estimator fits a six-parameter model to the measured phase-A current waveform. The model captures the subtransient component, the DC offset, and the steady-state current — all simultaneously, in a two-pass decomposition. The key innovation in the model is a continuity constraint that eliminates the DC amplitude as a free parameter. This reduces the Jacobian condition number from over 2000 to under 20 — a 250-times improvement — making the LM algorithm converge reliably every time. The R-squared of the fit is 0.999. From the fitted steady-state current, we derive an updated pickup I_p^{adapt} that correctly reflects the actual IBR penetration level at that moment."

[1:20–2:00] Explain the coordination gate:
"Two safety mechanisms prevent the adaptation from doing harm. First, a four-component confidence score γ checks the fit quality, the Jacobian conditioning, parameter stability, and physical plausibility. If γ drops below 0.70, the relay holds

its previous settings. In all three validated fault cases, γ was between 0.89 and 0.93 — comfortably above the gate. Second, the bounded update law clips the pickup change to a maximum of 10% per cycle and verifies that the coordination time interval constraint $t_{\text{primary}} + \text{CTI} \leq t_{\text{backup}}$ is satisfied for all IBR penetration levels before committing the new setting. This is what the slide means by projected-gradient optimisation."

[2:00–2:40] Present the results:
"The results are in the table. Spurious pickups fall by 58%, from 38 to 16 — that is the security margin gain visible in the figure, where the adaptive pickup at 1.49 pu sits above the load transient zone that was triggering the fixed pickup at 1.20 pu. The coordination margin improves from 0.82 to 0.94. Most importantly, maloperation under IBR penetration drops from 12.3% to 1.1%. The framework maintains full IEC 60255-151 compliance throughout."

[2:40–3:00] Close:
"This is Contributions C6, C7, and C8. The LM model achieves R-squared of 0.999. The confidence gate correctly authorises adaptation in all validated cases. And the bounded update law eliminates maloperation without compromising the coordination margin. The extension to high-impedance faults and multi-generator bus topologies is planned as C9, and the Park *dq* truth model validation is planned as C10."

Anticipated Questions and Answers

Q1: The IEC Very-Inverse formula uses M squared. Why is the slide's formula using a general power B?

Answer: The slide presents the general IEC 60255-151 inverse-time family, which covers Standard Inverse ($A = 0.14$, $B = 0.02$), Very Inverse ($A = 13.5$, $B = 2$), and Extremely Inverse ($A = 80$, $B = 2$). The thesis uses the Very-Inverse curve throughout, so in all numerical results $B = 2$ and the denominator is $(M^2 - 1)$ exactly as in the IEC standard. The general form is written on the slide so that the framework is presented as applicable to any IEC curve type, not just Very-Inverse. Different generator applications use different curves: Extremely Inverse for generator feeders with large inrush, Standard Inverse for cable-fed systems.

Q2: How does the relay know k-ibr? Does it require communication from the IBR?

Answer: The relay does not require direct communication from the IBR. The LM estimator infers the effective k_{ibr} from the shape of the measured fault current waveform itself — specifically, from the ratio of the steady-state current I_{ss} to the pre-fault load current I_{load}. A lower I_{ss} relative to I_{load} indicates higher IBR suppression of fault current. Alternatively, the EKF digital twin in Ch 4 estimates k_{ibr} directly from the terminal voltage and current phasors, and passes it to the SyncOC adaptation layer. Both paths are available; the chapter uses the LM-internal estimate (two-pass I_{ss}) as the primary path.

Q3: The DC continuity constraint eliminates one parameter. Why does this reduce the condition number so dramatically?

Answer: The seven-parameter model has two parameters describing the DC component: $I_{DC}(t_0)$ and τ_{dc}. These two parameters are strongly correlated: different combinations of (I_{DC}, τ_{dc}) can produce almost identical waveforms in the estimation window. This collinearity shows up as a near-singular row in the Jacobian, giving $\kappa_n \approx$ 2000–5800. The continuity constraint fixes $I_{DC}(t_0) = -I_{sub}\cos\phi_a - I_{ss}$, removing I_{DC} as a free parameter. The remaining six parameters describe physically distinct components with well-separated time scales ($\tau_{ac} \approx 30$ ms vs $\tau_{dc} \approx 100$ ms vs $1/\omega_0 \approx 3.3$ ms), so the Jacobian columns are nearly orthogonal. This orthogonality drops κ_n to 8–18.

Q4: Could the spurious pickup reduction cause a missed trip on a real high-impedance fault?

Answer: This is exactly the security concern that the bounds address. The adaptive pickup is bounded above by $I_{p,\max} = 0.80 \times I^{\min}_{\text{fault,3P}}$ — 80% of the minimum bolted three-phase fault current. This ensures that no matter how high the adaptive pickup is set, it always remains below the minimum real fault current. For SLG faults at $R_F = 50\,\Omega$ (high-impedance), the fault current is typically 1.3–2.0 pu at $k_{\text{ibr}} = 0.3$; the adaptive pickup never exceeds 80% of 2.86 pu = 2.29 pu. The 1.49 pu adaptive pickup shown in the paper is well below this limit. The table shows 0 missed trips on bolted faults and the same missed trip rate on SLG faults as the fixed relay (extension to HIF is planned C9).

Q5: How does this connect to WAPC? Isn't OC protection a local function?

Answer: Locally, yes — each relay makes its own trip decision independently. But the CTI constraint ($t_{\text{primary}} + 0.3 \leq t_{\text{backup}}$) is a coordination constraint between neighbouring relays. When SAMBP-SyncOC adapts its TMS, it changes its operating time, which affects whether it still coordinates correctly with the backup relay upstream. The WAPC (Ch 11) maintains a real-time CTI feasibility check across all relays in the network. It broadcasts updated CTI margin bounds to each relay as k_{ibr} changes across the network. SAMBP-SyncOC reads its local CTI bound from the WAPC before applying any setting change. If the WAPC is unavailable (communication failure), the relay uses the most conservative pre-stored CTI bound, reverting toward the commissioning-time settings rather than applying an unchecked adaptation.

Chapter 4 One-Line Summary for Examiners

"IBR depresses fault current by 30–60%, making fixed OC settings either too insensitive (missed trips) or too sensitive (spurious pickups). SAMBP-SyncOC fits a six-parameter waveform model using two-pass LM estimation ($R^2 = 0.999$, $\kappa_n < 20$), derives an adaptive pickup $I_{\text{set}}(k_{\text{ibr}})$, gates the update with a four-component confidence score ($\gamma \geq 0.70$), and projects onto the CTI feasibility set. Result: spurious pickups -58%, maloperation $12.3\% \rightarrow 1.1\%$, coordination margin $0.82 \rightarrow 0.94$."